

A numeric method of finding zeros of $\mathbb{R}^n \mapsto \mathbb{R}$ functions

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Let there be a function $f : \mathbb{R}^n \mapsto \mathbb{R}$, whose zeros are to be approximated. Let us consider the space $\mathbb{R}^{n+1} \ni (x_1, x_2, \dots, x_n, z)$. Constructing the iterative algorithm, which is, in a way, a generalization of the Newton-Raphson method, we consider a sequence of approximations $(p_i)_{i \in \mathbb{N}} \in (\mathbb{R}^n)^\infty$ of a zero of the function f , which is convergent to such zero. p_0 is to be given as input. Applying a heuristic, we consider a line

$$s : (x_1, x_2, \dots, x_n) = p_i - \lambda \nabla f(p_i) \quad \text{for } \lambda \in \mathbb{R}. \quad (1)$$

Approximating the values of the function in the neighbourhood of point p_i as $dz = \sum_{j=1}^n \frac{df}{dx_j} dx_j$, we get

$$z = z_0 + \sum_{j=1}^n \left((x_j - (x_0)_j) \left(\frac{df}{dx_j} \right)_{x_0} \right), \quad (2)$$

which for the line s comes down to

$$z = f(p_i) + \sum_{j=1}^n \left(\left((p_i)_j - \lambda \left(\frac{df}{dx_j} \right)_{p_i} - (p_i)_j \right) \left(\frac{df}{dx_j} \right)_{p_i} \right) = f(p_i) - \lambda \sum_{j=1}^n \left(\frac{df}{dx_j} \right)_{p_i}^2. \quad (3)$$

Finding the zero of this approximation, we obtain

$$\lambda = \frac{f(p_i)}{\sum_{j=1}^n \left(\frac{df}{dx_j} \right)_{p_i}^2}. \quad (4)$$

Therefore we have achieved the recursive formula

$$p_{i+1} = p_i - \frac{f(p_i)}{\sum_{j=1}^n \left(\frac{df}{dx_j} \right)_{p_i}^2} \nabla f(p_i). \quad (5)$$

It can easily be checked that is equivalent to the Newton-Raphson method for $n = 1$. This formula can also be written more concisely in the following way

$$p_{i+1} = p_i - \frac{f(p_i)}{\|\nabla f(p_i)\|^2} \nabla f(p_i). \quad (6)$$